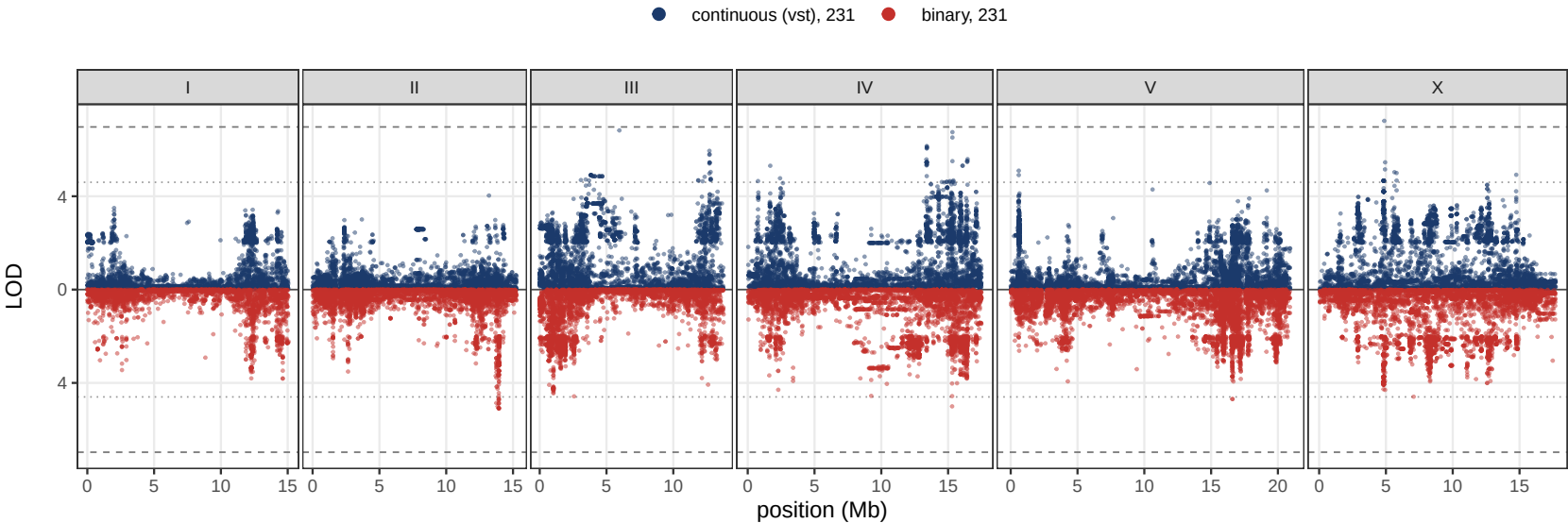


A

The same 231 strains, scored continuously (above) and as responsive / not (below)

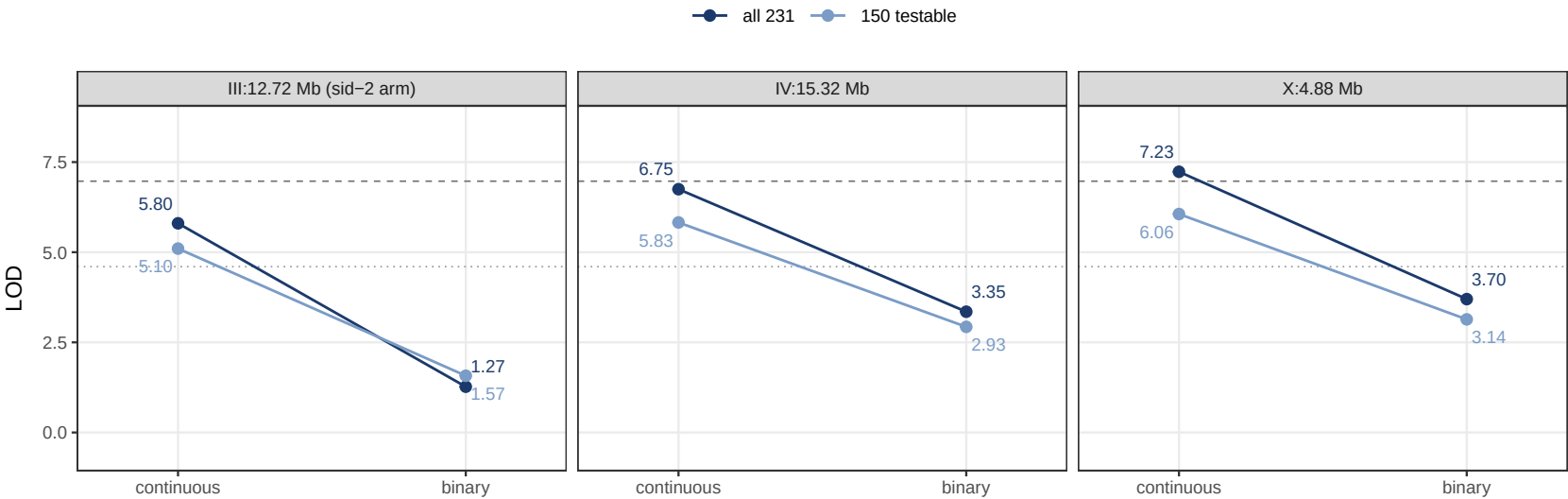
Dashed = Bonferroni 6.97, dotted = eigen 4.60. Continuous clears the eigen threshold at 101 markers and Bonferroni at 1; binary clears eigen at 18 and Bonferroni at 0. Every chromosome's top marker changes.



B

Crossing the scale against the strain set separates the two effects

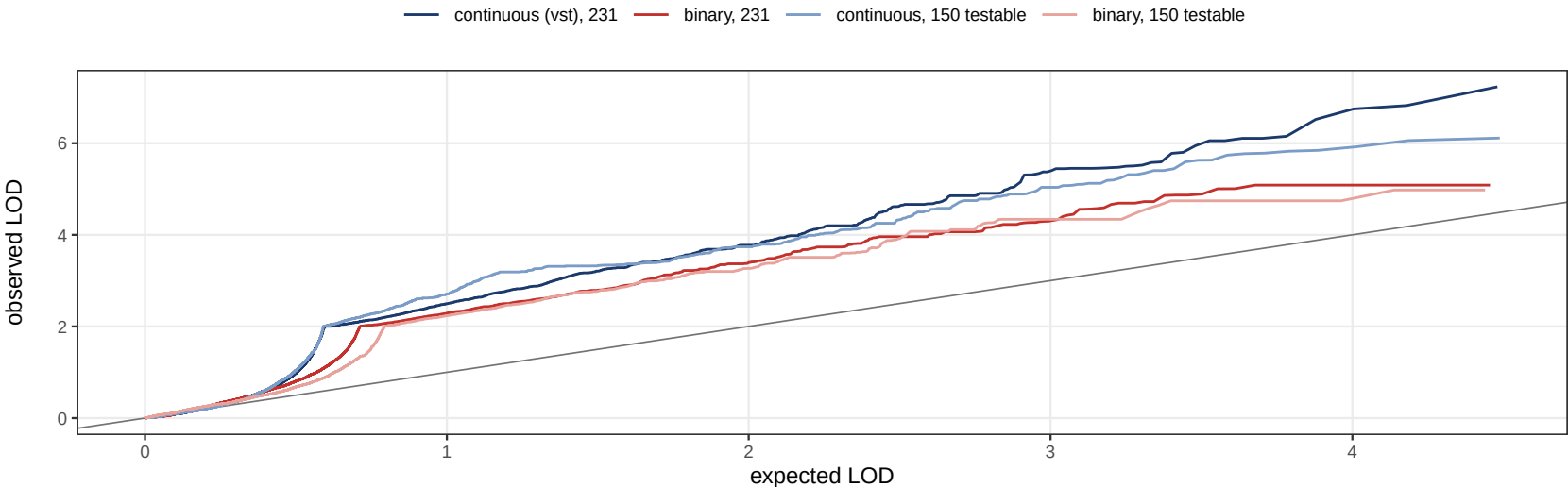
Holding the strain set fixed, continuous → binary is the steep drop. Holding the scale fixed, removing the 81 censored strains costs about one LOD -- so those strains are informative, and collapsing magnitude is what loses the signal. The chromosome III panel is the right-arm sid-2 locus III:12718465, not the spurious III:5965738 singleton that tops the scan.



C

The continuous scan is deflated; the binary scan is calibrated but flat

lambda_gc: continuous 0.631 and 0.517, binary 0.937 and 1.008. The continuous scans sit BELOW the null line, so the mixed model is over-correcting and the published p values are conservative rather than anti-conservative. Binarising fixes the calibration and removes the signal along with it.



D

On the vst scale the censored strains are not the most extreme -- but that is the scale's doing

A strain absent from every pos-1 pool has a LESS extreme vst value than one that declined sharply from a high control frequency, because absolute change is bounded by where it started. That is a property of the absolute scale, NOT evidence that the strains responded less: on the ratio scale, 80 of the 81 have a censoring bound below the median observed log2(pos/ctrl), so they do sit at the responsive end. Binarising still loses the mapping, but because it discards magnitude, not order.

